

Tutorial Quiz 10 Rubric

Q1]

Stokes' Theorem:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$$

To compute the curl:

$$P = y^2 + z^2 ; Q = x^2 + y^2 , Z = x^2 + y^2$$

$$(\nabla \times \mathbf{F})_x = R_y - Q_z = 2y - 0 = 2y$$

$$(\nabla \times \mathbf{F})_y = P_z - R_x = 2z - 2x$$

$$(\nabla \times \mathbf{F})_z = Q_x - P_y = 2x - 2y$$

$$\therefore (\nabla \times \mathbf{F}) = \langle 2y, 2z - 2x, 2x - 2y \rangle$$

On our surface, S , $z = 0$

$$\therefore (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{k}} = 2x - 2y$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_{-1}^1 \int_{-1}^1 (2x - 2y) dx dy$$

$$= \int_{-1}^1 -4y \, dy$$

$$= 0$$

$$\therefore \text{circulation } C = 0 \quad \text{---(1)}$$

$$\text{Q2]} \quad F = \sqrt{x^2 + y^2 + z^2} (x\hat{i} + y\hat{j} + z\hat{k})$$

$$D = \text{Spherical shell } 1 \leq x^2 + y^2 + z^2 \leq 2$$

* Solving this problem with cartesian coordinates can get messy and not recommended. However, final answers will be evaluated.

Using spherical coordinates simplifies the problem highly.

$$\text{Consider } F = r \langle x, y, z \rangle$$

$$\text{with } r^2 = x^2 + y^2 + z^2$$

Divergence :

$$\nabla \cdot F = \frac{\partial}{\partial x} (xr) + \frac{\partial}{\partial y} (yr) + \frac{\partial}{\partial z} (zr)$$

$$\begin{aligned}
&= \left(r + \frac{x^2}{r}\right) + \left(r + \frac{y^2}{r}\right) + \left(r + \frac{z^2}{r}\right) \\
&= 3r + \frac{1}{r} (x^2 + y^2 + z^2) \\
&= 3r + \frac{1}{r} \cdot r^2 = 4r
\end{aligned}$$

Divergence Theorem:

$$\iint_{\partial D} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_D 4r \, dV$$

In spherical coordinates

$$dV = r^2 \sin \theta \, dr \, d\theta \, d\phi$$

$$\text{with } r \in [1, \sqrt{2}], \theta \in [0, \pi]$$

$$\phi \in [0, 2\pi]$$

$$\iiint_D 5r^2 \, dV = 4 \int_1^{\sqrt{2}} \int_0^{\pi} \int_0^{2\pi} r (r^2 \sin \theta) \, d\phi \, d\theta \, dr$$

$$= 4 \int_1^{\sqrt{2}} r^3 \, dr \int_0^{\pi} \sin \theta \, d\theta \int_0^{2\pi} d\phi$$

$$= 4 (2\pi) \cdot 2 \int_1^{\sqrt{2}} r^3 \, dr$$

$$= 16 \pi \left[\frac{r^4}{4} \right]_1^{\sqrt{2}}$$

$$= 4 \pi [4 - 1]$$

$$= 12 \pi$$

— (2)